Timbre and Similarity Judgments

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Abstract

Timbre remains an elusive auditory attribute. Many timbre researchers have incorporated human timbre similarity judgments into their studies. Often, pairwise timbre similarity judgments have been averaged and then analyzed to create timbre spaces, attempted maps of human perceptions of timbre. This paper offers a select chronology of such timbre similarity research, from traditional psychoacoustics methods to a more recent ecological approach. I contrast psychoacoustic experiment designs with those of perceptual similarity research in psychology and offer considerations for future timbre similarity research.

1 Introduction

Timbre remains an elusive auditory attribute after decades of study. In their attempts to model and establish the physical correlates of timbre, researchers since the 1960s have adopted the methodology of collecting human similarity judgments. Often, pairwise timbre similarity judgments have been averaged and analyzed to create timbre spaces, attempted maps of human perceptions of timbre. This paper offers a select history of such timbre similarity research, from these traditional psychoacoustic methods to a more recent ecological approach. The paper contrasts psychoacoustic experiment designs involving timbre similarity judgments with the experiment designs of perceptual similarity researchers in psychology. It also offers possibilities for future timbre similarity research.

Section 2 offers definitions and background information. Section 3 presents a chronological selection of timbre similarity studies from 1977 through 2020, moving from a traditional psychoacoustics paradigm to an ecological paradigm. Section 4 outlines research on similarity judgment from the field of psychology. This psychological research presents alternative experiment designs and findings about the perceived similarity of multidimensional stimuli which might inform timbre similarity research. Section 5 lists research questions suggested by the previous sections.

2 Background

2.1 Defining Timbre

While perceived audio attributes like loudness, pitch, and duration have measurable physical correlates, there is not just one physical quantity that can be varied to induce a change in the perception of timbre. Timbre perception seems tied to a large and loose collection of sonic elements, including measurable audio features as well as associations present in the hearer's prior experience which may or may not be present in the audio signal. Some of the relevant audio attributes are global properties of a sound and some are time-varying properties with widely varying time scales. Different individuals weight each of these elements differently when perceiving timbre. Defining timbre psychoacoustically is complex and has therefore been a subject of research for over sixty years.

2.2 Measuring Timbre Similarity

To get a handle on timbre perception, researchers often ask participants to judge the similarities of audio stimuli. The designs of the stimuli, the interfaces used for collecting participant judgments,

and the mathematical techniques used to turn these judgments into representations of timbre vary across research teams and reflect a variety of motivations and assumptions about timbre.

2.3 Timbre Space

One common representation of timbre adopted by researchers is that of a multidimensional *timbre space*. Grey's 1977 publication gives more detail about how a timbre space is created (Grey (1977)). A timbre space treats sound stimuli as points in a geometric space with an arbitrary number of dimensions (see Figure 1); the similarity of the stimuli is treated as a distance between stimuli in space, or a metric. The timbre space approach is motivated by the attempt to find measurable audio attributes that covary with each axis of the timbre space. An assumption built into the timbre space approach is that participants' timbre similarity judgments are consistent and satisfy the requirements of a metric. [1] This approach assumes that timbre can meaningfully be represented as a metric space (Hunter (2014)).

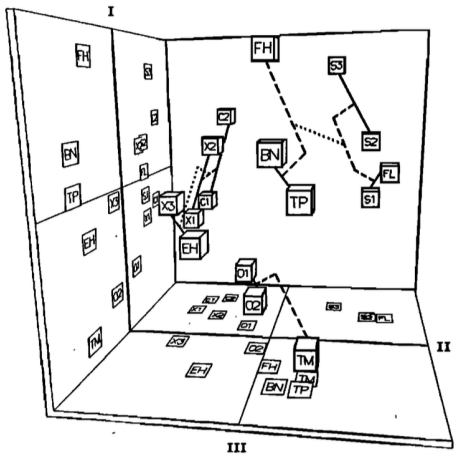


Figure 1: Grey's introduction of a three-dimensional timbre space.

2.4 Changing Approaches

As the previous decades of psychoacoustics research have proceeded, approaches have shifted from the search for universal physical correlates of auditory percepts to a more ecological approach that attempts to consider context, audio events, auditory streams, material causes of sound, and prior listener associations. The study of timbre has similarly grown from an attempt to find measurable audio attributes that universally determine the perception of timbre into a more context-aware approach that considers transmodal perception and situatedness (Neuhoff (2004)).

3 Select Studies of Timbre Similarity

Below, I present a selection of timbre similarity studies from 1977 through 2020. The laboratory studies from the 1970-1990s follow a traditional psychoacoustics paradigm, while the later studies increasingly adopt an ecological approach to timbre in context.

[1] A metric is another word for a distance. There are three requirements for a metric:

1. $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$. That is, distances are always positive.
2. $d(x, y) = d(y, x)$. That is, the distance from x to y is the same as the distance from y to x.
3. $d(x, y) \leq d(x, z) + d(z, y)$, the triangle inequality.

3.1 Grey (1977)

In 1977, Grey published an influential experiment attempting to identify the spectral attributes of sound most relevant for timbre perception (Grey (1977)). The methodology he established was followed by many later researchers. Grey synthesized imitations of 16 musical instrument timbres in order to have precise control over harmonic content, loudness, duration, and pitch. He gathered similarity judgments from participants on pairs of the 16 tones, averaged the judgments across participants, and performed multidimensional scaling and clustering on the resulting similarity matrix. The result was a map of the tones in a three-dimensional timbre space (see Figure 1). Grey attempted a physical interpretation of the three axes of timbre space, labeling them *spectral energy distribution*, *spectral fluctuation*, and *temporal patterning*. He found that the order of presentation of the tones affected their judged similarity but that multiple sessions with the same participant yielded the same judgments. Grey's 1977 experiment forms the basis of much ensuing timbre similarity research.

3.2 McAdams et al. (1995); McAdams (1999)

Stephen McAdams and his team at IRCAM have conducted research on timbre for over thirty years (McAdams et al. (1995); McAdams (1999, 2012, 2019)). Their work in the 1990s largely adopts Grey's 1977 methodology, using synthesized instrument tones, multidimensional scaling, and clustering algorithms to convert participants' pairwise similarity judgments into a timbre space whose axes are physically interpretable. In 1995, McAdams' team noted that participants in these studies fell into five latent classes of listeners, with each class emphasizing different sonic attributes in their assessments of timbre similarity. The five listener classes were not divided along biographical or professional lines. By the late 1990s, the team had expanded their research into studying timbre intervals, identifying more specific physical correlates for the axes of timbre space (*log attack time*, *spectral centroid*, *spectral smoothness*, and *spectral flux*), and gauging how broader or narrower sonic contexts affected timbre similarity judgments. McAdams and team found that timbre similarity judgments vary in amplitude in a changed context but that their relative proportions remained the same, implying a robustness of timbre similarity judgment to context change. The team also identified *specificities*, sonic attributes specific to a certain instrument or sound source that tended to inform timbre similarity judgments more than other audio attributes. McAdams and team continue to influence the direction of timbre research.

3.3 Toiviainen et al. (1995, 1998)

In parallel with the 1990s work of McAdams at IRCAM, Peter Toiviainen et al. were computationally modeling timbre spaces using early artificial neural networks (Toiviainen et al. (1995, 1998)). They gathered similarity judgment data using a methodology similar to Grey's. They additionally fed their audio samples through a computational model based on the human auditory system and a Kohonen self-organizing map (SOM), training it for 50,000 iterations. They had the model issue its own similarity judgments and compared them to the human judgments, and they found that the model had a moderate degree of success at modeling the human judgments. They concluded that the inter-subjective correlations of 0.65-0.9 in similarity judgments and the ability of the model to somewhat recreate those judgments implied that timbre could appropriately be modeled as a metric space and timbre similarity as a metric. They also found that a SOM trained on raw audio rather than audio preprocessed with the human auditory model was significantly worse at replicating human judgments, emphasizing the value of considering how the human auditory system processes sound in data engineering for machine learning audio tasks.

3.4 Peeters et al. (2011)

In 2011, a team led by Peeters released the Timbre Toolbox, a Matlab library that extracts a large set of audio features relevant for timbre research (Peeters et al. (2011)). The development of the Timbre Toolbox involved Peeters et al.'s experimentation with a variety of audio input representations and an extensive analysis of the intercorrelation of both global and time-varying audio features. The team noted that in prior research, time-varying properties of audio signals had been represented with a global average, and they called for new time series representations of time-varying audio attributes. They also recommended a method for constructing small, orthogonal sets of audio features for use in timbre similarity research.

3.5 Siedenburg and Müllensiefen (2017)

In the 2010s, Siedenburg and Müllensiefen studied timbre similarity judgments of music recordings rather than synthesized tones, a move toward an ecological approach to psychoacoustics (Siedenburg and Müllensiefen (2017); Siedenburg and McAdams (2018)). The researchers selected equal-length instrumental excerpts of music tracks from four genres, and they asked participants to group them into four groups of four items by overall similarity. They employed linear regression to find the most relevant physical attributes for the perception of similarity. They also included track metadata as possible predictive attributes. Siedenburg and Müllensiefen found that the most significant correlate of similarity judgment was the genre metadata descriptor, with participants' perception of genre seeming to take precedence over any audio attributes for similarity judgment. They suggested that further studies of musical timbre should include not only audio attributes but also music metadata. They also suggested including a third anchor stimulus in timbre comparisons rather than pairwise similarity rating.

3.6 McAdams (2019)

McAdams' approach to timbre has evolved since his earlier work, and as of the 2020s, he has embraced an ecological approach to timbre perception (McAdams (2019)). In 2019, he wrote, "Timbre is a perceptual property, not a physical one." He notes that humans are highly sensitive to tiny spectrotemporal microstructures in sound and that it's hard to fully account for these sensitivities in a laboratory setting. He notes that almost all prior studies have treated time-varying properties of a sound by some kind of averaging and, like Peeters, calls for research that treats time-varying properties as time series. He notes that timbral brightness, at least, has been shown to fulfill the requirements of a ratio scale; otherwise, he de-emphasizes the idea of timbre space or physical correlates of timbre. His current research centers on listener experience of materials and actions that guide perceptions of timbre and sound events.

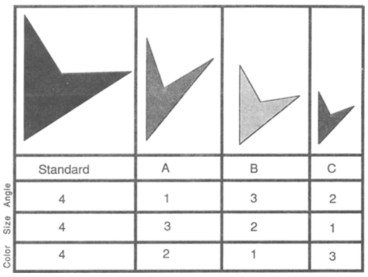
4 Similarity Judgment Outside Psychoacoustics

Researchers outside the field of psychoacoustics have researched similarity judgments of complex stimuli, as well. The following visual studies from the field of psychology use strategies that diverge from the experiment design adopted by Grey and subsequent timbre researchers. These psychology studies may suggest alternative approaches to the study of timbre similarity.

4.1 Goldstone et al. (1997)

Goldstone et al. carried out a 1997 series of experiments on the effects of context on similarity judgment (Goldstone et al. (1997)). They intended to challenge researchers' assumptions about the reliability of similarity judgments, and they argued against the idea of multidimensional space as a representation of complex perceptual phenomena. In their first experiment, they found that human similarity judgment of complex stimuli violated transitivity. A person presented with three complex visual stimuli and a reference would judge B to be more similar to the reference than A, C more similar than B, and A more similar than C (see Figure 2, left). Two different varieties of intransitivity arose in their results, and participants seemed to shift the dimensions to which they paid the most attention within a single trial. Goldstone et al.'s second experiment presented participants with a reference face and groups of additional faces, each of which shared some of the reference face's features (see Figure 2, right). They found no clear, shared strategy for participants' similarity judgments, and they classified the participants into a few categories based on their varying responses to the stimuli. They noted that the attention drawn by a specific feature during similarity judgments correlated with its increased variability within the overall context. They asserted that averaging across the entire population of participants would be meaningless because participants' judgments fitted a multimodal distribution. The third experiment explored the idea that a stimulus may spontaneously bring to mind external stimuli that then affect similarity judgments. The third experiment did not offer conclusive results, but it did find that participants chose to background different features of a complex stimulus based on the context and even the location of the visual on a page. The team concluded that "dimensional salience is modified on line, at the time of a comparison" and that "the properties that are used for evaluating similarity are fully fixed only after the comparison process has begun" Goldstone et al. (1997). The more values a stimulus takes on

along a certain dimension, the more importance is accorded to that dimension. They emphasized that this does not mean any stimulus can be made to seem arbitrarily similar to any other stimulus, but they did note the inevitability of context effects on similarity judgments.



	Standard	A	B	C
Angle	4	1	3	2
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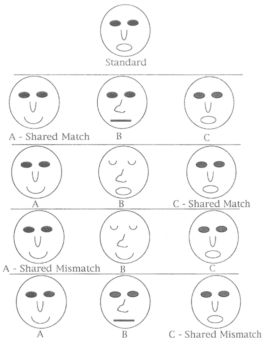


Figure 2: Stimuli for similarity judgments from Goldstone's Experiment 1 (left) and Experiment 2 (right).

4.2 Hepting et al. (2019)

In 2019, Hepting et al. conducted an experiment studying how participants grouped similar complex visual stimuli (Hepting et al. (2019)). The team presented participants with a stack of 356 photographs of faces and asked the participants to sort the photos into any number of piles, each pile containing similar faces. The participants were only allowed to see the topmost face from each pile at one time, making each sorting action a series of pairwise similarity judgments. Of the hundreds of photo pairs studied, zero pairs were unanimously rated as similar, while 232 pairs were unanimously related as dissimilar. Because there was no ground truth against which to measure the similarity judgments, the research primarily resulted in a framework of similarity judgment with three classes (similar, dissimilar, and undecided). The team was unable to discern a common strategy by which these similarity judgments were made across participants.

5 Suggested Research Questions in Timbre Similarity

Each of the studies described above points to possible future research. Grey's 1977 finding that the order of tone presentation matters for similarity judgments dovetails with Goldstone's 1997 findings that similarity judgments of multidimensional visual stimuli are intransitive and that the location of a stimulus on a page matters for similarity judgments. Grey's and Goldstone's results together suggest that it could be valuable to recreate Goldstone's transitivity and context experiments with timbres varying along perceptually relevant dimensions. A finding that timbre similarity judgments are intransitive would conflict with the representation of timbre as a timbre space and the assumption that timbre similarity judgments satisfy the requirements of a metric (Hunter (2014)). Multiple researchers have already performed experiments confirming the metric nature of certain audio attributes relevant for timbre or have claimed the propriety of timbre space as a representation (McAdams (1999); Kazazis et al. (2022); Toiviainen et al. (1995)), but a finding of intransitivity of timbre similarity judgments would complicate this picture. McAdams' 1990s findings that listeners fall into latent classes combines with Hepting's methods in studying facial similarity to suggest that it could be valuable to have participants freely group timbral stimuli and assess whether distinct latent classes of timbre hearers can be discerned. If so, their distinct timbre perception strategies could be studied (McAdams (1999); Hepting et al. (2019)). Toiviainen's finding that neural networks perform much better at reproducing human similarity judgments when the input audio is processed through an intricate model of the human auditory system suggests that future AI models could benefit from having their training data and inputs pre-processed by perceptual models (Toiviainen et al. (1995)). Siedenburg's suggestion that timbre research should employ three-way comparison rather than pairwise comparison echoes Goldstone's finding that the total context of a similarity judgment matters (Siedenburg and Müllensiefen (2017)). Peeters and McAdams both call for new time series representations of time-varying audio attributes (Peeters et al. (2011); McAdams (2019)). As

psychoacoustics embraces a more ecological approach, McAdams', Siedenburg's, and Goldstone's methods imply that the consideration of more realistic audio stimuli and total participant context could be fertile ground for future timbre research (McAdams (2019); Siedenburg and Müllensiefen (2017); Goldstone et al. (1997)).

6 Conclusion

Although timbre has been studied for decades by an engaged research community, more areas for exploration remain. Timbre may or may not be best represented as a metric space, and timbre similarity judgments may or may not behave like a metric. Listeners may or may not fall into distinct latent classes that differently prioritize sonic elements in an algorithmically replicable way. Contextual and musical associations carried only indirectly by an audio signal may have as much to do with timbre perception as attributes extractable from an audio signal. As machine learning gains presence in the fields of psychoacoustics and music, a better understanding of timbre could contribute to making successful models that mirror human perceptive and creative processes. Timbre similarity judgments are yet to be fully understood, modeled, or imitated.

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